What is computable?

David Hilbert

1900 — Vision for 20th century

1. Cantor's problem of the cardinal number of the continuum.
2. The compatibility of the arithmetical axioms.
3. Scissor congruence of polyhedra of equal volumes.
4. Problem of the straight line as the shortest distance between two points.
5. Lie's concept of a continuous group of transformations without the assumption of differentiability of the functions defining the group.
6. Mathematical treatment of the axioms of physics.
7. Irrationality and transcendence of certain numbers.
8. Problems of prime numbers.
9. Proof of the most general law of reciprocity in any number field.
10. Determination of the solvability of a Diophantine equation.
11. Quadratic forms with any algebraic numerical coefficients.
12. Extensions of Kronecker's theorem on Abelian fields to any algebraic realm of rationality.
13. Impossibility of the solution of the general equation of 7th degree by means of functions of only two arguments.

⋮

1928 — Entscheidunsproblem

give an algorithm that, given any logical proposition, determines whether it is T or F

Alan Turing

1936 — gives formal def. of computation (TM)

- Showed that no TM exists to check "Does M halt on w"
- TMs can run other TMs

1950s onward:

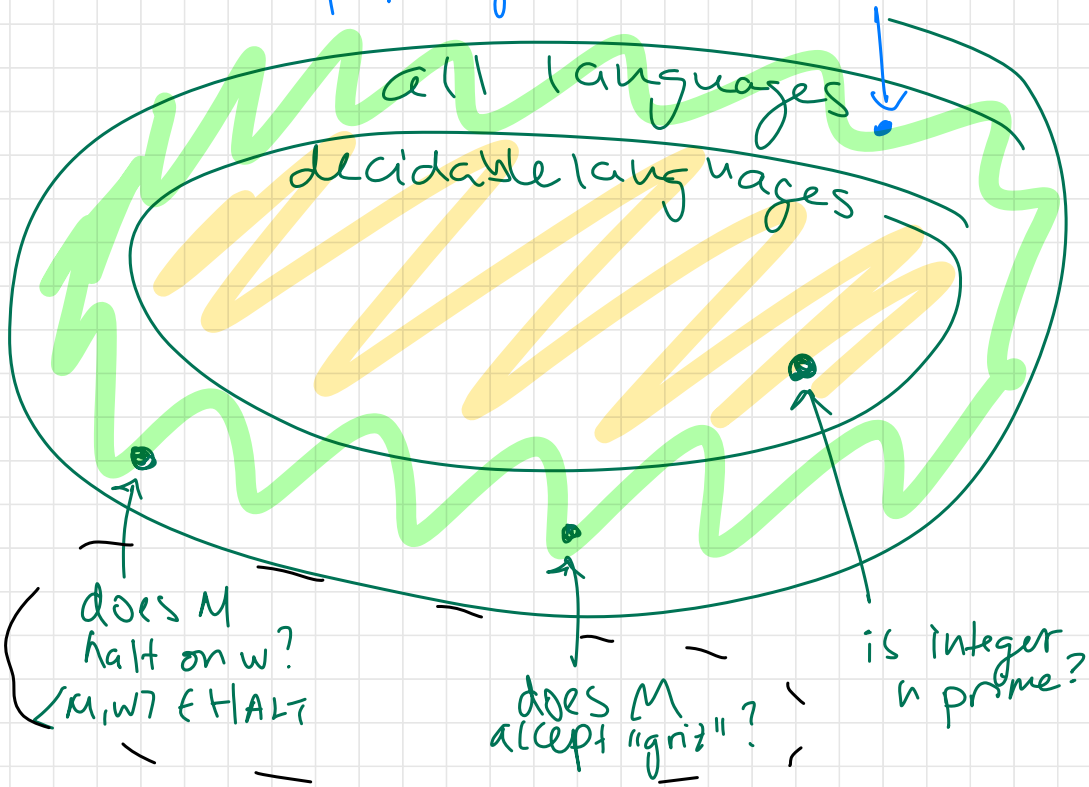
- regular expressions + finite automata (Kleene)
- grammars (Chomsky)
- computation as membership in sets of strings

$w \in L$

does x have property Y?

$5x^2 + 3y + z = 0$

does a polynomial equation have integer sol's?



Most decision problems about behavior of TMs are undecidable.

Diagram not to scale!

$|S|$ $\quad |\mathcal{P}(S)| = 2^{|S|}$

all languages using alphabet Σ is uncountably infinite $\quad \Sigma = \mathcal{P}(\Sigma^*)$

set of all subsets

all decidable languages = all (encodings of) TMs = set of strings is countably infinite

Diagonalization proofs use this

Another use for reduction

Let A be a decidable problem.

How fast is the fastest alg. to solve A?

1. Give an alg. for A (upper bound)

ex give n^2 alg — A can be solved no slower than n^2

2. Reduce problem B (that we know something about) to A (lower bound)

give an alg. to solve B using A

ex B can't be solved faster than $n \log n$

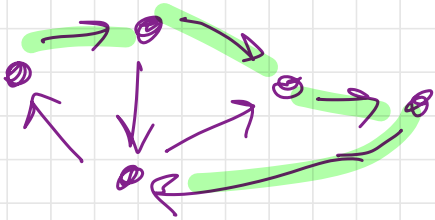
reduce B to A — A can't be solved faster than $n \log n$

Let LongPath(G, L) be the problem of deciding whether G has a path of length L.:

simple

does not repeat nodes

# of edges

$G =$ 

$L = 4$ T

$L = 5$ F

(TM)

Suppose you have a magic black box that decides LongPath(G, L).

The problem MaxPathLength =

input: G

output: length of longest path in G

reduce MaxPathLength to LongPath.